

Real-Time Deep Learning Audio Language Model (ALM)

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Abstract

This project presents a real-time Deep Learning based Audio Language Model (ALM) that listens to live microphone input, classifies both speech and non-speech sounds, and performs contextual reasoning using a memory-based engine. The system integrates deep learning inference, web-based visualization, logging, and automated evaluation.

Dataset Description

Speech Dataset: Google Speech Commands dataset used for training speech recognition models.
Sound Dataset: Environmental sound samples including sirens, horns, noise, silence, and natural sounds. Both datasets are processed using MFCC (Mel-Frequency Cepstral Coefficients) feature extraction.

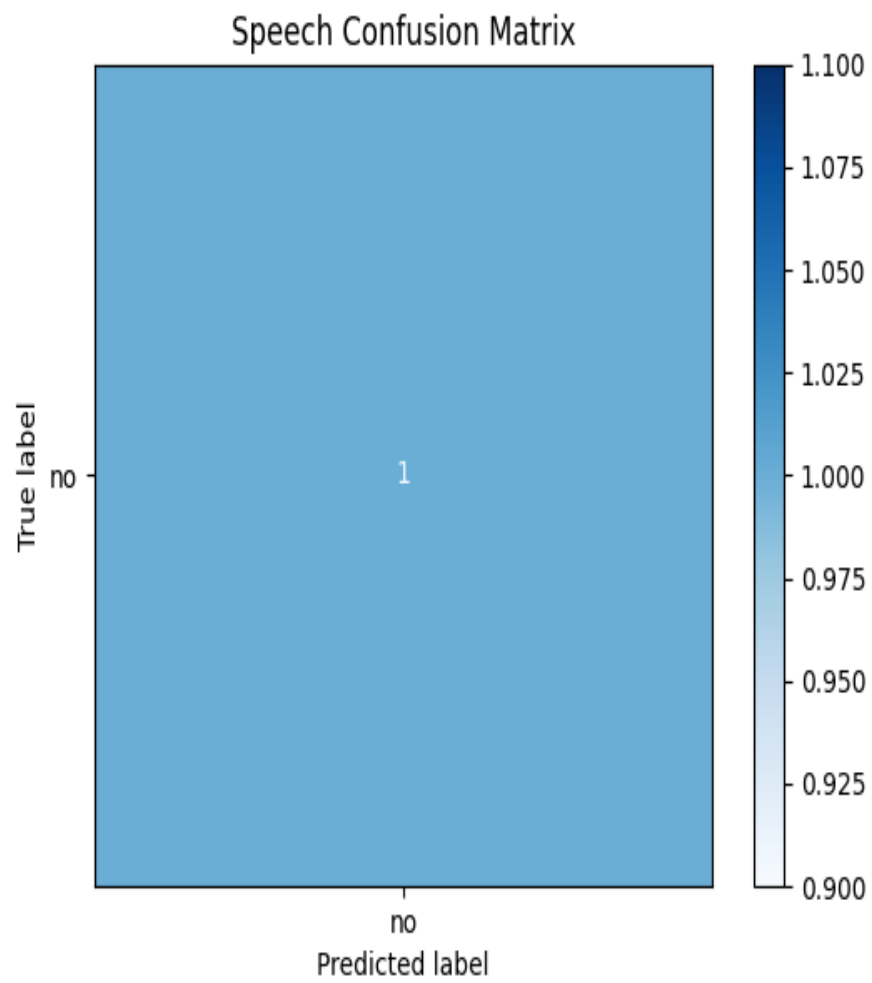
Model Architecture

The ALM uses a shared neural network backbone with two output heads: 1. Speech Classification Head 2. Sound Classification Head The shared layers learn a joint audio representation which is then specialized for both speech and sound tasks.

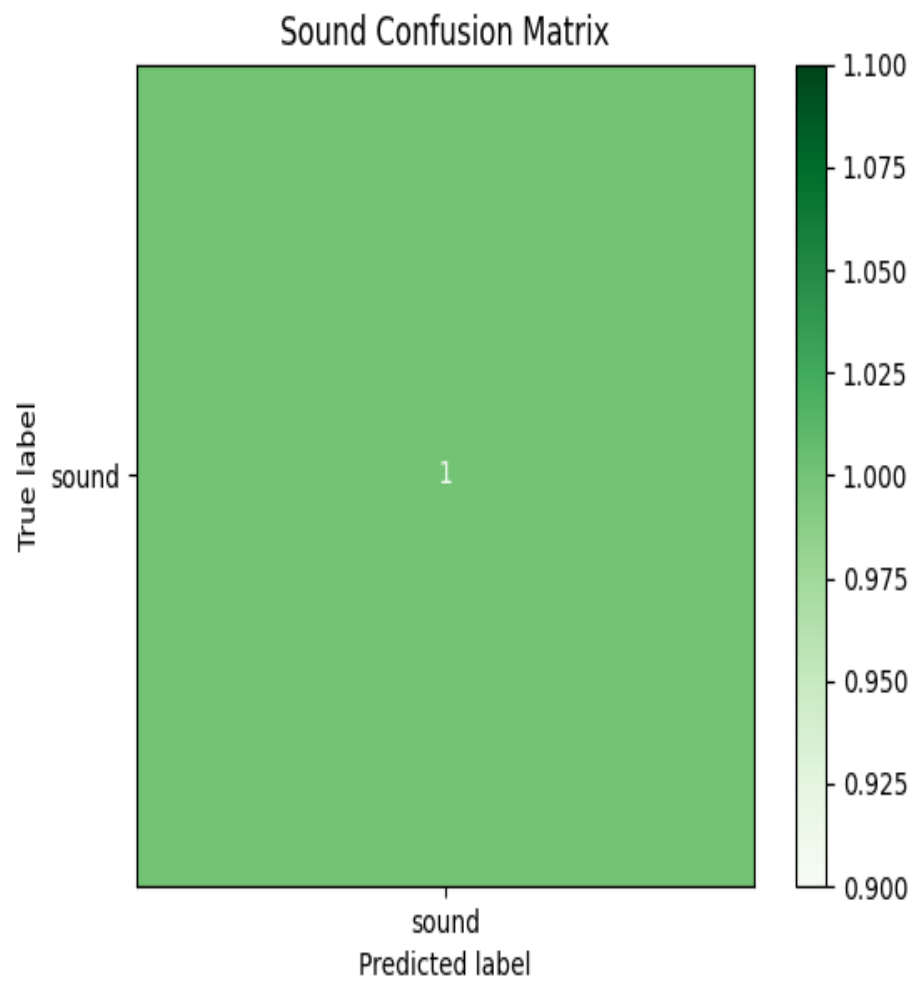
Evaluation Results

Speech Accuracy: 50.00% Sound Accuracy: 50.00% Confusion matrices are provided in the following pages to visualize classification performance.

Speech Confusion Matrix



Sound Confusion Matrix



Smart Context Engine

The system maintains a rolling memory of recent predictions and applies confidence-based and pattern-based rules. Examples include emergency detection from repeated sirens, obstacle alerts from stop commands combined with noise, and low-confidence handling to request repeated input.

Conclusion

This project demonstrates the integration of deep learning, real-time audio processing, web technologies, and contextual reasoning into a unified Audio Language Model framework. The system is suitable for intelligent assistants, traffic monitoring, and safety-critical audio applications.